VOLODYMYR (VLAD) TKACHUK

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EDUCATION

2027 **PhD in Computing Science** (expected) **– University of Alberta**
Advisors: Csaba Szepesvári & Xiaoqi Tan

2023 **MSc in Computing Science – University of Alberta**
Advisor: Csaba Szepesvári

2021 **BSc in Electrical Engineering – University of Waterloo** (91.9% GPA in final two years)

TEACHING EXPERIENCE

2025 **Instructor** (Fall term)
University of Alberta – CMPUT 267 (Machine Learning I) [website] [notes] [lecture videos]
310 students, 12 TAs. Second time teaching. Added practice exams and extra exercises in response to student feedback from Fall 2024.

2025 **Head TA** (Winter term)
University of Alberta – CMPUT 267 (Machine Learning I) [website]
308 students, led 10 TAs; course used my materials from Fall 2024.

2024 **Instructor** (Fall term)
University of Alberta – CMPUT 267 (Machine Learning I) [website] [notes] [lecture videos]
334 students, 11 TAs. Redesigned the course and wrote all course materials: 166-page course notes (used in place of a textbook), lectures, theory assignments, autograded Colab notebooks, and exams. Materials adopted by later instructors (Winter 2025, Fall 2026).

2024 **Head TA** (Winter term)
University of Alberta – CMPUT 267 (Basics of Machine Learning) [website]
208 students, led 8 TAs.

2023 **Guest Lecturer** (8 lectures) and **TA** (Winter term)
University of Alberta – CMPUT 605 (Graduate course: Theoretical Foundations of RL) [website]

2022 **TA** (Winter term)
University of Alberta – CMPUT 653 (Graduate course: Theoretical Foundations of RL) [website]

2021 **TA** (Fall term)
University of Alberta – CMPUT 396 (Intermediate Machine Learning) [website]

TEACHING AWARDS

2025 Graduate Student Teaching Award (University of Alberta)

2021 Graduate Student Teaching Award (University of Alberta)

PUBLICATIONS

[1] **Gaussian-Width Bounds for Offline-to-Online Linear Bandits**
Kushagra Chandak, Volodymyr Tkachuk, Xiaoqi Tan, Csaba Szepesvári (Preprint)

[2] **Trajectory Data Suffices for Statistically Efficient Policy Evaluation in Fixed-Horizon Offline RL with Linear q^{π}-Realizability and Concentrability**
Volodymyr Tkachuk, Csaba Szepesvári, Xiaoqi Tan (COLT 2026)

[3] **Trajectory Data Suffices for Statistically Efficient Learning in Offline RL with q^{π}-Realizability and Concentrability**
Volodymyr Tkachuk, Gellért Weisz, Csaba Szepesvári (NeurIPS 2024)

[4] **Regret Minimization via Saddle Point Optimization**
Johannes Kirschner, Alireza Bakhtiari, Kushagra Chandak, Volodymyr Tkachuk, Csaba Szepesvári (NeurIPS 2023)

[5] **Efficient Planning in Combinatorial Action Spaces with Applications to Cooperative Multi-Agent Reinforcement Learning**
Volodymyr Tkachuk, Seyed Alireza Bakhtiari, Johannes Kirschner, Matej Jusup, Ilija Bogunovic, Csaba Szepesvári (AISTATS 2023)

[6] **Investigating action encodings in recurrent neural networks in reinforcement learning**
Matthew Kyle Schlegel, Volodymyr Tkachuk, Adam M. White, Martha White (TMLR 2023)

[7] **On Efficient Planning in Large Action Spaces with Applications to Cooperative Multi-Agent Reinforcement Learning**
Volodymyr Tkachuk (MSc Thesis)

PUBLIC TALKS

2026 **Trajectory Data Suffices for Statistically Efficient Policy Evaluation in Fixed-Horizon Offline RL with Linear q^{π}-Realizability and Concentrability**
(IFDS Seminar, University of Washington, 2026) | (Amii AI Seminar, 2025)

2025 **Trajectory Data Suffices for Statistically Efficient Learning in Offline RL with q^{π}-Realizability and Concentrability**
(RL Theory Seminars, 2025) | (Amii AI Seminar, 2024)

SCHOLARSHIPS AND FELLOWSHIPS

2026 President's Doctoral Prize of Distinction (**$7,100)**

2025 NSERC Postgraduate Scholarship – Doctoral (PGS D) (**$40,000/year x 3 years)**

2025 President's Doctoral Prize of Distinction (**$10,000)**

2024 Alberta Graduate Excellence Scholarship **($12,000)**

2021 NSERC Canada Graduate Scholarship – Master's (CGS M) (**$17,500)**

2021 Walter H Johns Graduate Fellowship **($5,800)**

2020 Gerry Heckman Scholarship **($3,000)**

2018 Dan and Anik Colquhoun Award **($2,000)**

RESEARCH EXPERIENCE

2026 **Visiting Researcher**
University of Washington – Advised by Kevin Jamieson

2021 **Research Assistant**
University of Alberta – Advised by Martha White

2020 **Research Assistant**
University of Alberta – Advised by Matthew E. Taylor

2019 **Research Engineering Intern**
Apple – Acoustics Special Projects Group

PROFESSIONAL EXPERIENCE

2019 **Audio System Tuning Intern**
Apple – Telephony Tuning Team

2018 **Advanced CMOS Sensor Development Intern**
Teledyne DALSA – CMOS Sensor Products Team

2017 **Digital Tool Developer and Verification Engineering Intern**
Peraso Technologies – Digital Verification Team

2016 **Web Developer**
University of Waterloo – Undergraduate Recruitment Group

2016 **Software Engineering Intern**
T. Hong & Co. – Web Development Team

SERVICE

2027 AISTATS Reviewer

2025 NeurIPS Reviewer

2023 Reinforcement Learning (RL) Theory seminar co-host/organizer [website]